

TaxaID: A Modular R Ecosystem for Bayesian Taxonomic Assignment

Project Overview and Purpose

Biologists increasingly measure biodiversity from sequence, sound, and image data, yet current pipelines can have high false-positive and false-negative rates for taxonomic assignment. TaxaID is a modular ecosystem of nine R packages that improve taxonomic assignment accuracy. A typical input is a table of candidate matches for each observation (sequence, image, or acoustic recording) previously obtained by querying a reference library. TaxaID can implement traditional score-threshold approaches, but its main advances are to:

1. screen for reference-database errors,
2. detect taxa missing from the reference library,
3. consider whether a taxon is plausible at the sampling location, and
4. apply Bayes' Theorem to generate assignment probabilities from match scores.

To apply Bayes' Theorem, TaxaID converts match scores to likelihoods, estimates spatially explicit occurrence-based priors, and computes posterior probabilities of taxonomic identity. The ecosystem was designed with eDNA metabarcoding in mind, but image and acoustic analyses are possible when starting from a table of candidate matches.

The ecosystem supports three workflows, ranging from simple to comprehensive:

- **Traditional workflow** – Select a consensus taxon using score thresholds and/or a lowest common ancestor.
- **LLM-shortcut workflow** – Use large language models (Anthropic Claude, Google Gemini, OpenAI, or local Ollama) to rapidly estimate priors and generate consensus assignments.
- **Bayesian workflow** – Train a likelihood model on reference data, build spatially explicit priors from GBIF occurrences, and compute posteriors via Monte Carlo simulation.

These workflows converge at the same posterior consensus step, enabling direct comparison of model-based and LLM-based assignments.

Many TaxaID functions can use large language models (LLMs), though most have non-LLM alternatives. LLM integration requires an API key (see below).

Common Errors in Taxonomic Assignment

Automated classifiers for DNA, sound, and image data produce taxonomic assignments with systematic errors that are often difficult to detect. In a replicated eDNA study from the California rocky intertidal, 28% of metazoan sequences matched taxa not present on the Pacific Coast (Shea and Boehm 2024). Camera trap false positive rates can exceed 40% for rare species (Thompson et al. 2025), and acoustic classifier precision is highly sensitive to confidence threshold settings (Fairbairn et al. 2025). Raw classifier scores mimic probabilities but are uncalibrated — a 0.95 score does not mean 95% confidence (Dussert et al. 2025), and the same match percentage can be diagnostic for one taxon group but ambiguous for another (Ficetola et al. 2015). These errors fall into three categories: false positives (wrong taxon assigned), false negatives (correct taxon missed), and combined errors where one taxon's false positive is another's false negative.

Reference quality **Reference mislabeling** (FP) — Misabeled sequences or images in the reference database produce confident wrong assignments that propagate to every query matching that reference. *TaxaLikely detects mislabels before model training and removes them from match data.*

Missing reference redirect (FP + FN) — When the true species is absent from the reference library, its detections are assigned to the closest relative — a false positive for that relative and a false negative for the true species. Reference library gaps are geographically biased, systematically affecting some regions and taxa more than others (Marques et al. 2021). *TaxaLikely models the expected score profile of unreferenced taxa, and TaxaAssign identifies and names plausible missing species.*

Score interpretation **Overconfident species assignment** (FP + FN) — Uncalibrated raw scores are taken at face value; even a 100% match may be ambiguous at species rank if competing candidates score nearly as well. *TaxaLikely’s calibrated likelihoods reveal that a high score with a small gap to alternatives has low species-level likelihood, regardless of the raw score.*

Overly strict thresholds (FN) — Conservative score cutoffs discard correct assignments that fall just below arbitrary thresholds. *TaxaID’s probabilistic framework replaces binary thresholds with continuous likelihoods and posteriors.*

Ecological context **Defensive upranking** (FN) — When multiple similar species produce near-identical scores, conventional systems uprank to genus to avoid a false positive, sacrificing species-level resolution. *Spatial priors from TaxaExpect can break ties: if only one candidate species is expected at the site, its posterior can support species-level assignment even when scores alone cannot. Dynamic Bayesian updating further sharpens priors within a sample after high-confidence detections.*

Ecologically implausible assignment (FP) — A species is assigned that doesn’t plausibly occur at the sampling location, season, or habitat. *Spatially explicit priors from TaxaExpect down-weight implausible taxa. Dynamic updating within a sample reinforces ecologically consistent assignments.*

Field and lab artifacts **Contamination or artifact** (FP) — Lab or field contamination, handler artifacts (camera traps), or equipment carryover introduces real detections of taxa not present in the environment. *TaxaFlag detects proportion-based and temporal-proximity artifacts.*

Allochthonous transport (FP) — Detections originate from outside the sampling area: eDNA carried by runoff or currents, sounds from playback devices or captive animals. *Spatial priors inherently down-weight species outside their expected habitat.*

Additionally, TaxaFlag provides LLM-based expert review that can flag most of these error types post-assignment.

Ecosystem Packages

1. **TaxaTools** cleans and standardizes taxonomic names across backbones (GBIF, NCBI, WoRMS, Catalogue of Life) and provides a unified interface for calling LLMs from R.
2. **TaxaFetch** acquires species occurrence records from GBIF, DataONE, BioTIME, and published literature.
3. **TaxaHabitat** classifies taxa into habitat categories using LLM-based biological consensus and flags spatial outliers.
4. **TaxaMatch** standardizes match tables from external tools (BLAST, camera-trap classifiers, acoustic detectors) into a common format.
5. **TaxaLikely** converts match scores into calibrated likelihoods using a hierarchical Bayesian model trained on the reference library. It also audits references for mislabels and coverage gaps.
6. **TaxaExpect** builds spatially explicit Bayesian priors by modeling species occurrence probability across a geographic grid, incorporating habitat and spatial autocorrelation.

7. **TaxaAssign** computes posterior probabilities from likelihoods and priors, generates consensus taxonomy, and produces publication-ready reports.
8. **TaxaFlag** flags anomalous detections after assignment: contamination (lab/field blanks), handler artifacts (camera traps), and ecologically implausible assignments.
9. **TaxaWizard** interviews the user about their data and goals, then generates a complete R script, methods section, or Shiny application.

Dependency Chain

```
TaxaTools -> TaxaFetch -> TaxaHabitat -> TaxaExpect -> TaxaAssign -> TaxaFlag
TaxaMatch -> TaxaLikely -> TaxaAssign -> TaxaFlag
TaxaWizard (standalone; generates scripts that call the other packages)
```

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Licensing

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See LICENSE.md for details. TaxaExpect depends on glmmTMB (GPL ≥ 3); the TaxaExpect source code itself is CC0, but binary distributions that bundle glmmTMB may be subject to GPL terms.

Related Software

Several R packages and standalone tools address taxonomic assignment from DNA barcoding data. TaxaID differs from these in its explicit separation of likelihood and prior, spatially explicit priors built from occurrence data, and multi-data-type support (DNA, image, acoustic).

Table 1. Comparison of TaxaID with related taxonomic assignment tools.

Tool	Approach	Posterior probabilities	Spatial priors	Unreferenced taxa	Multi-data-type
TaxaID	Generative Bayesian (MVN score + gap)	Yes (full distribution)	Yes (GBIF + habitat)	Yes (NCBI census + LLM)	Yes

Tool	Approach	Posterior probabilities	Spatial priors	Unreferenced taxa	Multi-data-type
PROTAX (Somervuo et al. 2017)	Bayesian with taxonomy-tree prior	Yes	No	Yes (tree-based)	No (DNA only)
BayesANT (Zito et al. 2023)	Bayesian nonparametric (kmer)	Yes	No	Yes (Pitman-Yor)	No (DNA only)
IdTaxa / DECIPHER (Murali et al. 2018)	Phylogenetic ML	Bootstrap confidence	No	No	No (DNA only)
insect (Wilkinson et al. 2018)	Profile HMM + classification tree	Akaike weights	No	No	No (DNA only)
SINTAX (Edgar 2016)	Kmer + bootstrap	Bootstrap confidence	No	No	No (DNA only)
RDP Classifier (Wang et al. 2007)	Naive Bayes (8-mer)	Bootstrap confidence	No	No	No (DNA only)
DADA2 assignTaxonomy (Callahan et al. 2016)	Naive Bayes (kmer)	Bootstrap confidence	No	No	No (DNA only)
galaxy-tool-lca (Beentjes et al. 2019)	Score-threshold + LCA (deterministic)	No	No	No (upranked)	No (BLAST/DNA only)

TaxaID is complementary to several of these tools rather than a replacement. DADA2 or OBITools handle upstream sequence processing; TaxaMatch ingests their output. DECIPHER is used internally by TaxaLikely for reference sequence alignment. The key innovation of TaxaID is that the same likelihood model can be combined with different spatial priors at different sites, and that the framework extends to image and acoustic data via TaxaMatch score standardization.

The galaxy-tool-lca tool (Beentjes et al. 2019; <https://github.com/naturalis/galaxy-tool-lca>) is a widely used Python tool for LCA-based taxonomic assignment from BLAST results, particularly for freshwater macroinvertebrate eDNA and fungal ITS metabarcoding. Its core algorithm — filtering BLAST hits by identity, bitscore, and query coverage, then finding the lowest common ancestor among passing hits — is conceptually similar to TaxaID’s `score_consensus()`. A notable strength of galaxy-tool-lca is its explicit use of **query coverage** (the fraction of the query sequence that aligns to each reference hit) as a mandatory quality filter, alongside percent identity and bitscore. A 98% identity match that covers only half the amplicon is weaker evidence than one spanning the full amplicon, and coverage is already available as a BLAST output column (`qcovs`). TaxaID’s likelihood model currently uses score (percent identity) and the gap to the second-best candidate as its primary signals but does not incorporate alignment coverage. Users can partially address this upstream by filtering on coverage in TaxaMatch before passing match data to TaxaLikely; incorporating coverage as a third dimension in the likelihood model is a potential future enhancement. The analogous quality signal for acoustic reference data is the Xeno-canto quality grade (A–E per recording), which `TaxaLikely::fetch_reference_recordings()` already uses to filter reference recordings before model training.

Table 1b. Comparison of TaxaID (image path) with standalone image classifiers. TaxaID converts raw classifier confidence scores into calibrated likelihoods via a reference training set with known species identity, then multiplies those likelihoods by spatially explicit priors; the tools below produce the raw scores that TaxaMatch and TaxaLikely process.

Tool	Taxa scope	Score output	Spatial priors	Unreferenced taxa	R access
TaxaID (image path)	Any (classifier-agnostic)	Calibrated likelihoods $\rightarrow$ Bayesian posteriors	Yes (GBIF + habitat)	Yes (coverage audit)	Native
animl / SpeciesNet (Tabak et al. 2019; Wildlife Insights)	Camera trap mammals	Confidence (0–1), top-5; rollup ensemble	No	No	animl R package
iNaturalist computer vision	General wildlife (108,000+ taxa)	Softmax (0–1), top-10; genus/family fallback	Limited (app UI only)	No	rinat (indirect)
InsectNet (He et al. 2025)	Insects (2,526 spp, 17 orders)	Conformal prediction sets; OOD energy score	No	Limited (OOD flag)	Web app only
Seek / iNaturalist mobile	General wildlife	Community consensus	No	No	None
Wildlife Insights (WCS/Google)	Camera trap wildlife	Confidence (0–1), rollup ensemble	No	No	None (web platform)

TaxaID is downstream of, not competing with, these classifiers. The key point is that raw confidence scores from neural networks are uncalibrated softmax outputs — a 90% confidence score does not mean a 90% chance the identification is correct. TaxaLikely addresses this by fitting a generative model to a labeled reference set of images with known species identity, capturing the full score distribution for correct matches (H1), wrong-species matches (H2), and absent-species responses (H3). Animl results are read directly by `TaxaMatch::read_animl_output()`; iNaturalist CV JSON output is read by `TaxaMatch::read_inaturalist_cv_output()`; Wildlife Insights / SpeciesNet batch predictions are read by `TaxaMatch::read_wildlife_insights_output()`. For acoustic and image data, TaxaLikely acts as a post-classifier calibration layer: classifier output is standardized to `match_df` by TaxaMatch, then `unreferenced_candidates() + assign_scores()` convert classifier confidence scores to likelihoods (no separate reference-building step required).

InsectNet (He et al. 2025) is a notable recent advance for invertebrate specialists. Its two-stage self-supervised pretraining pipeline — general visual features from 3.6 billion Instagram images, followed by domain-specific pretraining on 12 million unlabeled insect images, then supervised fine-tuning on iNaturalist labels — achieves 96.4% top-1 accuracy across 2,526 species in 17 insect orders. The standout methodological innovation is replacing point confidence scores with **conformal prediction sets**: rather than returning a single species estimate, the model returns a set of candidate species guaranteed to contain the true species with 97.5% probability, with the set expanding as uncertainty grows. An energy-based out-of-distribution (OOD) score flags images outside the training distribution before classification is attempted. The conformal guarantee is conceptually related to TaxaID’s H1/H2/H3 framework — the true species is either in the candidate set (H1/H2) or flagged as OOD (H3 analog) — but the outputs are not directly compatible with `train_likelihood_model()` without access to the underlying softmax scores, which are not currently exposed via a public API. InsectNet also supports global-to-local fine-tuning with as few as $k=10$ images per new species, making regional deployment feasible for under-represented invertebrate groups. A web interface is available at <https://insectapp.las.iastate.edu>.

For contamination detection, the R package `decontam` (Davis et al. 2018) uses DNA concentration and prevalence to identify contaminants at the ASV level before taxonomic assignment. `TaxaFlag` operates after assignment, flagging implausible taxonomic identities using proportion-based control comparison, temporal proximity analysis, and LLM expert review.

Data and Hardware Requirements

- **Internet access** is required for GBIF queries, NCBI BLAST, and LLM API calls. Offline operation is possible when using cached data and local Ollama models.
- **No specialized hardware** is required. All packages run on standard desktop hardware (macOS, Linux, or Windows) with R $\geq 4.1.0$.
- **Input data** varies by entry point:
 - *eDNA workflow*: DADA2 sequence table or FASTA file, plus sample metadata.
 - *Image/acoustic workflow*: match score table with taxonomic identifications.
 - *Name-only workflow*: a list of taxonomic names and site coordinates.

Software Requirements

Table 2. Software dependencies required for the `TaxaID` ecosystem.

Software	Version	OS bit	Reference
R	$\geq 4.1.0$	64	R Core Team. 2025. R: A Language and Environment for Statistical Computing. V.4.5.2. https://www.r-project.org .
Bioconductor (DECIPHER, Biostrings)	≥ 3.17	64	Gentleman et al. 2004. Bioconductor. https://www.bioconductor.org . Required only for <code>TaxaLikely::build_sequence_matrix()</code> .
rBLAST	≥ 0.99	64	Hahsler and Nagar. 2019. rBLAST. https://github.com/mhahsler/rBLAST . Optional; required only for local BLAST in <code>TaxaMatch</code> .

All other R package dependencies are declared in each package’s DESCRIPTION file and will be installed automatically by `devtools::install()` or `install.packages()`.

API Keys

To access LLM tools, the user can use a locally installed LLM (Ollama) or an API key provided by an LLM service (set in `~/Renvirom`). A user could use a different LLM by modifying one of the existing LLM calls.

Key	Required By	How to Obtain
ANTHROPIC_API_KEY	TaxaTools (LLM calls)	https://console.anthropic.com/
GEMINI_API_KEY	TaxaTools (LLM calls)	https://aistudio.google.com/apikey (free tier)
OPENAI_API_KEY	TaxaTools (LLM calls)	https://platform.openai.com/ (paid)
OPENALEX_API_KEY	TaxaFetch (literature search)	https://openalex.org/settings/api (free)

No API key is needed for GBIF or NCBI queries. At least one LLM provider key is needed for the LLM-shortcut workflow, TaxaHabitat habitat assignment, and TaxaWizard.

Software Installation

```
# Install from GitHub
install.packages("remotes")

packages <- c("TaxaTools", "TaxaFetch", "TaxaHabitat", "TaxaMatch",
             "TaxaLikely", "TaxaExpect", "TaxaAssign", "TaxaFlag",
             "TaxaWizard")

for (pkg in packages) {
  remotes::install_github("kdlaafferty/TaxaID", subdir = pkg)
}

# Bioconductor dependency (needed for TaxaLikely reference matrix building)
if (!requireNamespace("BiocManager", quietly = TRUE))
  install.packages("BiocManager")
BiocManager::install("DECIPHER")
```

After installation, load the packages:

```
library(TaxaTools)      # Name cleaning, LLM providers (auto-detects API keys)
library(TaxaFetch)      # Occurrence data acquisition
library(TaxaHabitat)    # Habitat assignment
library(TaxaMatch)      # Match standardization
library(TaxaLikely)     # Score-to-likelihood conversion
library(TaxaExpect)     # Spatially explicit priors
library(TaxaAssign)     # Posterior computation and consensus
library(TaxaFlag)       # Post-assignment quality flagging
library(TaxaWizard)     # Interactive workflow designer
```

Not all packages are needed for every workflow. At minimum, load TaxaTools (always required) plus the packages for your workflow.

Instructions for Using Software

Packages

Package	Purpose	License
TaxaTools	Name verification, cleaning, parsing, rank lookup, column standardization; LLM provider functions; GBIF backbone census	CC0
TaxaFetch	Occurrence data acquisition (GBIF, DataONE, PDF, literature search), source combination	CC0
TaxaHabitat	Habitat assignment via LLM biological consensus; spatial QAQC	CC0
TaxaMatch	Match standardization. Sequence input (DADA2/FASTA), BLAST search.	CC0
TaxaLikely	Convert match scores to likelihoods using hierarchical Bayesian model; reference QC	CC0
TaxaExpect	Spatially explicit Bayesian priors from occurrence and habitat data	CC0
TaxaAssign	Posterior computation, consensus taxonomy, report generation	CC0
TaxaFlag	Post-assignment anomalous detection flagging (contamination, transport, scope)	CC0
TaxaWizard	Conversational LLM workflow designer: guided interview to .R script, .md methods, or Shiny app	CC0

Pipeline Overview

The diagram below shows the full set of supported workflow paths. Every path converges at the assignment step; users can start from whichever inputs they have.

```
flowchart TD
    %% Field data inputs → match_df
    F1["seqtab / FASTA\n(eDNA / metabarcoding)"] -->|TaxaMatch| MD
    F2["BirdNET CSVs\n(acoustic)"] -->|read_birdnet_output| MD
    F3["camera trap classifier output\nAniml · iNat CV · Wildlife Insights"] -->|read_animl/inat/wi_out| MD
    MD(["match_df"])

    %% Reference library building → model_params (DNA only)
    R1["taxa names"] -->|build_site_reference| RD
    R2["CRABS / local FASTA"] -->|"read_crabs_output  
read_reference_fasta"| RD
    RD(["reference_df"]) -->|build_sequence_matrix| RMAT(["DNA matrix"])
    RMAT -->|train_likelihood_model| MP(["model_params"])
```

```

%% Priors
TL["taxa + location"] -->|"TaxaFetch + TaxaHabitat
+ TaxaExpect"| PR(["prior_df"])

%% Likelihoods - three entry points
MD & MP -->|"compute_likelihoods\n(DNA: uses model_params)"| LK
MD -->|"assign_scores\n(acoustic / image: no model)"| LK
CON["consensus_df\n(morphology / expert IDs)"] -->|"unreferenced_candidates\n+ assign_scores"| LK
LK(["likelihoods"])

%% Assignment
LK -->|compute_posterior| PO
PR -->|compute_posterior| PO
PO(["posteriors"]) -->|"posterior_consensus
+ generate_report"| OUT["consensus assignment\n+ methods report"]

```

TaxaWizard can generate a complete R script for any of the paths above via a guided interview: `workflow_create()`.

Getting Started

Each package includes a vignette with a worked example:

```

# Ecosystem overview (recommended starting point)
vignette("taxaid-ecosystem", package = "TaxaAssign")

# Individual package vignettes
vignette("name-cleaning", package = "TaxaTools")
vignette("data-acquisition", package = "TaxaFetch")
vignette("habitat-assignment", package = "TaxaHabitat")
vignette("match-standardization", package = "TaxaMatch")
vignette("score-to-likelihood", package = "TaxaLikely")
vignette("building-priors", package = "TaxaExpect")
vignette("taxonomic-assignment", package = "TaxaAssign")
vignette("quality-flagging", package = "TaxaFlag")

```

Workflow Scripts

Detailed, runnable workflow scripts are provided in each package's `inst/` or `inst/workflows/` directory:

Workflow	Package	Script
FASTQ to match data (eDNA)	TaxaMatch	inst/workflow_fastq_to_match.R
Acoustic BirdNET workflow	TaxaMatch	inst/workflow_acoustics.R
Test: image reader functions	TaxaMatch	inst/test_image_functions.R
Fetch reference sequences (DNA)	TaxaLikely	inst/workflows/1_fetch_references_workflow
Flag reference errors	TaxaLikely	inst/workflows/2_flag_errors_workflow.R
Train likelihood model (DNA)	TaxaLikely	inst/workflows/3_train_model_workflow.R
Train model (acoustic)	TaxaLikely	inst/workflows/3b_acoustic_reference_workf
Train model (image)	TaxaLikely	inst/workflows/3c_image_reference_workflow
Score to likelihood	TaxaLikely	inst/workflows/4_score_to_likelihood_workf

Workflow	Package	Script
Audit reference coverage	TaxaLikely	inst/workflows/5_audit_coverage_workflow.R
Test: local reference library	TaxaLikely	inst/test_local_reference.R
Test: coverage calibration	TaxaLikely	inst/test_calibrate_coverage.R
Fetch occurrences	TaxaFetch	inst/Merge_sources_workflow.R
Assign habitats	TaxaHabitat	inst/Habitat_assign_workflow.R
Build priors	TaxaExpect	inst/TaxaExpect_workflow.R
LLM assignment	TaxaAssign	inst/TaxaAssign_llm_workflow.R
Bayesian assignment	TaxaAssign	inst/TaxaAssign_bayesian_workflow.R
Consensus	TaxaAssign	inst/TaxaAssign_consensus_workflow.R

For a step-by-step map of the full pipeline — inputs, outputs, and save conventions across all packages — see ECOSYSTEM_WORKFLOW.md.

High-Level Wrappers

For users who prefer a single-function interface:

```
library(TaxaAssign)

# Full Bayesian pipeline (~1 call, plus a saved model generated by TaxaLikely)
result <- run_bayesian_pipeline(match_df, model, site = list(lat = 34.4, lon = -119.8, main_habitat = "I"))

# LLM-shortcut pipeline (~1 call)
result <- run_llm_pipeline(match_df, site = list(lat = 34.4, lon = -119.8, main_habitat = "Marine"))
```

Interactive Workflow Designer

TaxaWizard provides a guided, conversational interface:

```
library(TaxaWizard)

# Opens a chat interface that interviews you about your data and goals,
# then generates a complete, runnable R script convertible to a shiny app.
workflow_create()
```

Data Outputs and Results

The TaxaID ecosystem produces outputs at each stage of the pipeline:

Occurrence Data and Habitat

- **Compiled occurrence records** from GBIF, DataONE, BioTIME, and literature extraction, standardized to Darwin Core columns (`stack_occurrences()`).
- **Habitat assignments** per taxon via weighted biological consensus across multiple habitat classification schemes (`assign_habitat_biological()`).
- **Spatial quality flags** identifying occurrences whose coordinates conflict with species habitat expectations (`flag_habitat_inconsistencies()`). These may be used to flag errant GBIF records.

Reference Library Assessment

- **Mislabel detection** identifying swapped or incorrectly labeled sequences in reference databases (`flag_reference_errors()`). This can help users avoid basing assignments on reference errors.
- **Coverage audits** enumerating described species per genus and flagging taxa absent from the reference library (`audit_barcode_coverage()`, `audit_reference_coverage()`). This can help users understand limits in precision.
- **Model diagnostics** summarizing expected match percentages, score gaps, and per-species profiles (`interpret_model()`). This can help users understand how scores relate to precision.

Species Distribution Models and Priors

- **Hierarchical biodiversity models** (glmmTMB) predicting species occurrence probability across a geographic grid as a function of habitat, spatial autocorrelation, and sampling effort (`train_biodiversity_model()`).
- **Spatially explicit Beta priors** (alpha, beta) for every taxon at every grid cell, including undetected diversity estimates for plausible but unobserved species (`generate_full_priors()`, `build_priors()`).
- **Interactive theta maps** – zoomable Leaflet maps showing predicted occurrence probability as color-coded grid heatmaps with occurrence points overlaid (`plot_theta_map_interactive()`).

Taxonomic Assignment

- **Calibrated likelihoods** from a hierarchical Bayesian model trained on reference-vs-reference match scores, supporting three hypothesis types: known species (H1), unreferenced species (H2), and unreferenced genus (H3) (`evaluate_likelihoods()`).
- **Posterior probabilities** of taxonomic identity for each observation, with Monte Carlo uncertainty estimates (`compute_posterior()`).
- **Consensus taxonomy** assignments at the finest rank supported by the data, with confidence scores and consensus method labels (`posterior_consensus()`, `score_consensus()`).

Quality Control and Reporting

- **Quality flags** for anomalous detections: contamination scores from lab/field blanks, temporal handler proximity, LLM expert review (`flag_contaminant()`, `flag_handler()`, `review_assignments()`).
- **Publication-ready text** (Methods and Results sections) via template-based and LLM-assisted report generation, with per-package report sections that assemble into a unified document (`generate_report()`, `assemble_report()`).

Software Inventory

Package	Exported Functions	Test Files	Vignette
TaxaTools	36	13	Yes
TaxaFetch	29	15	Yes
TaxaHabitat	13	5	Yes
TaxaMatch	10	7	Yes
TaxaLikely	22	14	Yes
TaxaExpect	13	9	Yes
TaxaAssign	14	13	Yes
TaxaFlag	4	4	Yes

Package	Exported Functions	Test Files	Vignette
TaxaWizard	7	3	Yes

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